NIFTY100 FINANCIAL INTELLIGENCE PLATFORM

Sprint 1 – Data Foundation

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Project Overview

Project Objective

- Build a robust data foundation for Nifty100 analytics.
- Create an ETL pipeline for financial datasets.
- Design a validated SQLite database.
- Perform data quality validation and testing.

Data Sources

Company Data

Companies

Documents

Pros & Cons

Financial Data

Profit & Loss

Balance Sheet

Cash Flow

Analytics

Analysis

Financial Ratios

Peer Groups

Market Data

Stock Prices

Sectors

Technology Stack

Category	Technology
Language	Python
Data Processing	Pandas
Database	SQLite
Testing	Pytest
Version Control	Git, GitHub
IDE	VS Code

ETL Architecture

Raw Excel Files

↓

Data Extraction

↓

Normalization

↓

Validation

↓

SQLite Database

↓

Reports & Audit Files

Database Design

Database Tables

- companies
- profitandloss
- balancesheet
- cashflow
- analysis
- documents
- prosandcons
- financial_ratios
- peer_groups
- sectors
- stock_prices

Data Quality Framework

DQ Checks Implemented

✓ DQ-01 Primary Key Validation

✓ DQ-02 Company-Year Validation

✓ DQ-03 Foreign Key Validation

✓ DQ-04 Null Value Analysis

✓ DQ-05 Negative Value Analysis

✓ DQ-06 Data Type Validation

✓ DQ-07 Range Validation

✓ DQ-08 Date Validation

Database Loading Results

Table	Records
companies	92
profitandloss	1276
balancesheet	1312
cashflow	1187
analysis	20
documents	1585
stock_prices	5520

All datasets were successfully loaded into the SQLite database without data loss.

Project Validation & Evidence

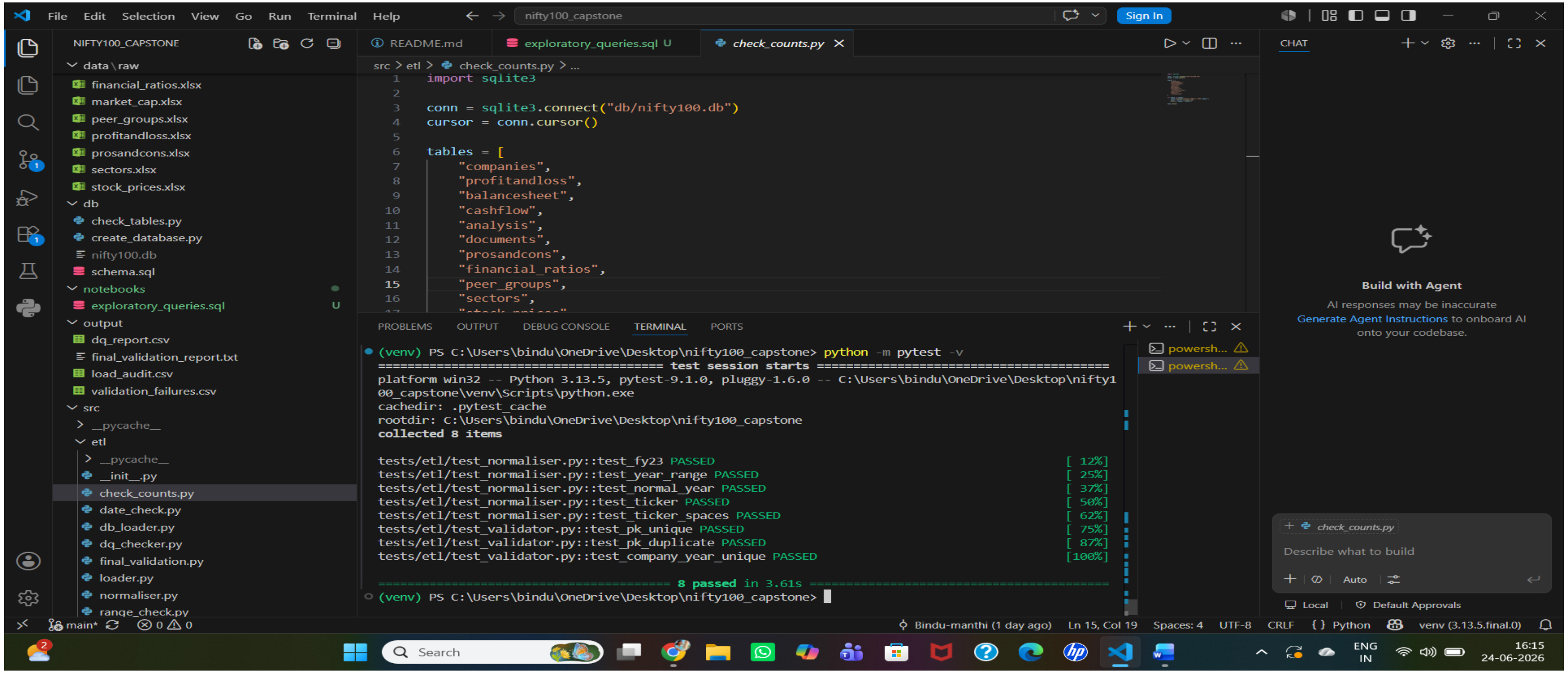

Figure 1: Successful execution of unit tests.

Project Validation & Evidence

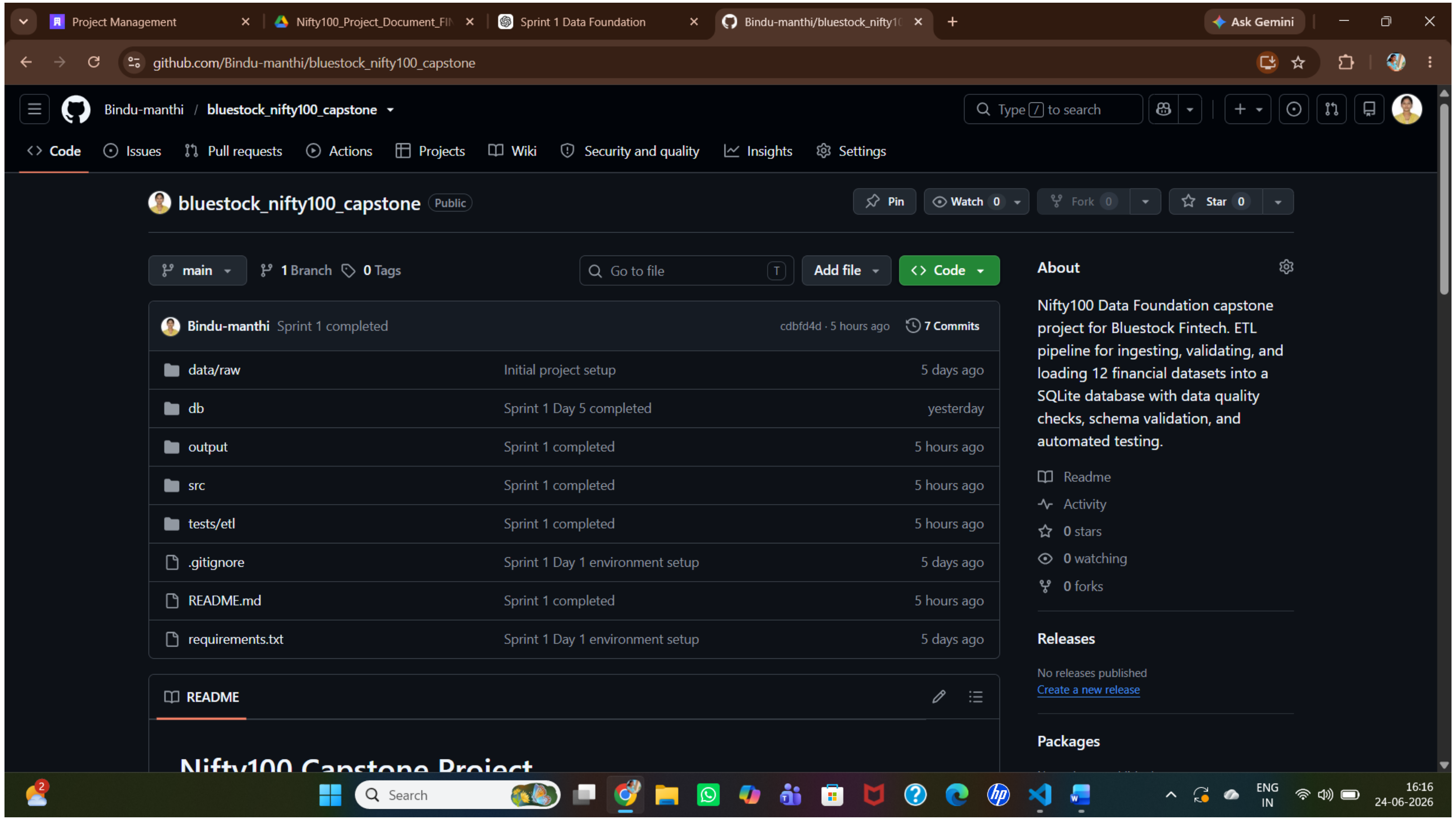

Figure 2: GitHub repository with project source code and documentation.

Conclusion

• Sprint 1 Completed Successfully

Achievements:

✓ ETL Pipeline Developed

✓ SQLite Database Created

✓ Data Quality Checks Completed

✓ Validation Reports Generated

✓ Project Documentation Completed

Ready for Future Analytics Modules

Project Status: Sprint 1 Completed Successfully ✅